

SEMESTER 2 EXAMINATIONS 2021/2022

MODULE:	CA208 - Logic
PROGRAMME(S):	CASE ECSAO
	BSc in Computer Applications (Sft.Eng.) Study Abroad (Engineering & Computing).
YEAR OF STUDY:	2,O
EXAMINERS:	Dr. David Sinclair (Internal) (Ext:5510)
TIME ALLOWED:	2 hours
INSTRUCTIONS:	Answer <u>ALL</u> 6 questions from Section A. Answer 2 out of 3 questions from Section B. Question 1 carries 30 marks. Questions 2 to 9 carry 10 marks.

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The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

Section A

QUESTION 1

[Total marks: 30]

Every child loves Santa. Everyone who loves Santa loves reindeer. Rudolph is a reindeer and Rudolph has a red nose. Anything that has a red nose is weird or a clown. No reindeer is a clown. Scrooge does not love anything which is weird.

1(a) [10 Marks]

Convert the statements above into a set of predicate formulae. *Hint:* You can use the following predicates: $child(x)$, $loves(x, y)$, $reindeer(x)$, $rednose(x)$, $weird(x)$, $clown(x)$.

1(b) [10 Marks]

Convert the predicate formulae from Question 1(a) into *Prenex Conjunctive Normal Form (PCNF)*.

1(c) [10 Marks]

Using *General Resolution* prove that Scrooge is not a child.

[End Question 1]

QUESTION 2

[Total marks: 10]

[10 Marks]

Without using truth tables, show that

$$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q.$$

Clearly indicate which logical equivalence is used at each step.

[End Question 2]

QUESTION 3

[Total marks: 10]

[10 Marks]

Using semantic tableaux, prove the following propositional formula is valid.

$$(p \rightarrow q) \rightarrow ((p \rightarrow \neg q) \rightarrow \neg p)$$

Clearly indicate which rule is used at each step.

[End Question 3]

QUESTION 4**[Total marks: 10]**

[10 Marks]

Using the Gentzen System $\mathcal{G}$ prove

$$\vdash (A \vee (B \wedge C)) \rightarrow ((A \vee B) \wedge (A \vee C))$$

Clearly indicate which rule is used at each step.

[End Question 4]**QUESTION 5****[Total marks: 10]**

[10 Marks]

Using *semantic tableaux* prove the following first-order statement is valid.

$$\vdash (\forall x.p(x) \vee \forall x.q(x)) \rightarrow \forall x.(p(x) \vee q(x))$$

Clearly indicate which rule is used at each step.

[End Question 5]**QUESTION 6****[Total marks: 10]**

[10 Marks]

Using *Skolemisation*, convert the following predicate expression into a set of clauses.

$$\exists y.(\exists x.R(x, y) \vee \exists x.S(x)) \rightarrow \neg(\neg\exists x.S(x) \wedge \forall x.\exists y.R(y, x))$$

Clearly indicate what is happening at each step.

[End Question 6]

Section B

QUESTION 7

[Total marks: 10]

[10 Marks]

Write Prolog relations to:

- sum the elements of a list of integers; and
- find the maximum of a list of positive integers.

[End Question 7]

QUESTION 8

[Total marks: 10]

[10 Marks]

Write a Prolog relation *split(N, Even, Odd)* that is true if all the even numbers (including 0) in the list *N* are in the list *Even*, and all the odd numbers in the list *N* are in the list *Odd*.

[End Question 8]

QUESTION 9

[Total marks: 10]

[10 Marks]

Give a detailed description, in your own words, of the procedural interpretation of a Prolog program.

[End Question 9]

APPENDICES

Semantic Tableaux

α	α_1	α_2	β	β_1	β_2
$\neg\neg A_1$	A_1		$\neg(B_1 \wedge B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \wedge A_2$	A_1	A_2	$B_1 \vee B_2$	B_1	B_2
$\neg(A_1 \vee A_2)$	$\neg A_1$	$\neg A_2$	$B_1 \rightarrow B_2$	$\neg B_1$	B_2
$\neg(A_1 \rightarrow A_2)$	A_1	$\neg A_2$	$B_1 \uparrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \uparrow A_2)$	A_1	A_2	$\neg(B_1 \downarrow B_2)$	B_1	B_2
$A_1 \downarrow A_2$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \leftrightarrow B_2)$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$A_1 \leftrightarrow A_2$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$	$B_1 \oplus B_2$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$\neg(A_1 \oplus A_2)$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\forall x.A(x)$	$A(a)$	$\exists x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$	$\neg\forall x.A(x)$	$\neg A(a)$

Gentzen Logic

$$\frac{\vdash U'_1 \cup \{\alpha_1, \alpha_2\}}{\vdash U'_1 \cup \{\alpha\}} \quad \frac{\vdash U'_1 \cup \{\beta_1\} \quad \vdash U'_2 \cup \{\beta_2\}}{\vdash U'_1 \cup U'_2 \cup \{\beta\}}$$

$$\frac{\vdash U'_1 \cup \{\gamma, \gamma(a)\}}{\vdash U'_1 \cup \{\gamma\}} \quad \frac{\vdash U'_1 \cup \{\delta(a)\}}{\vdash U'_1 \cup \{\delta\}}$$

$\frac{A}{B}$ is a *sequent* and means that if A is true, then B can be inferred.

α	α_1	α_2	β	β_1	β_2
$\neg\neg A$	A		$B_1 \wedge B_2$	B_1	B_2
$\neg(A_1 \wedge A_2)$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \vee B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \vee A_2$	A_1	A_2	$\neg(B_1 \rightarrow B_2)$	B_1	$\neg B_2$
$A_1 \rightarrow A_2$	$\neg A_1$	A_2	$\neg(B_1 \uparrow B_2)$	B_1	B_2
$A_1 \uparrow A_2$	$\neg A_1$	$\neg A_2$	$B_1 \downarrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \downarrow A_2)$	A_1	A_2	$B_1 \leftrightarrow B_2$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$\neg(A_1 \leftrightarrow A_2)$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$	$\neg(B_1 \oplus B_2)$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$A_1 \oplus A_2$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\exists x.A(x)$	$A(a)$	$\forall x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$	$\neg\exists x.A(x)$	$\neg A(a)$

[END OF APPENDICES]

[END OF EXAM]